



METABOLIC & GLP-1

MOTS-c

MOTS-c – 40 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Mitochondrial Open Reading Frame of the 12S rRNA-c
SYNONYMS / ALIASES	MOTS-c; mitochondrial-derived peptide (MDP)
CAS NUMBER	1627580-64-6
MOLECULAR FORMULA	C100H156N32O25S
MOLECULAR WEIGHT	Approximately 2174.6 g/mol
AMINO ACID SEQUENCE	MRWQEMGYIFYPRKLR (16 residues)
SOURCE	Synthetic (mitochondrially encoded native sequence; produced by solid-phase peptide synthesis)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	40 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Methionine residues are susceptible to oxidation; protect from prolonged air exposure. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

MOTS-c is a 16-amino-acid peptide encoded within the mitochondrial 12S ribosomal RNA region of the mitochondrial genome. It is classified as a mitochondrial-derived peptide (MDP). The native sequence contains a methionine at the N-terminus, two methionines internally, and a basic C-terminal region rich in arginine and lysine.

SELECTED PEER-REVIEWED REFERENCES

1. Lee C. et al. - *The mitochondrial-derived peptide MOTS-c promotes metabolic homeostasis and reduces obesity and insulin resistance* - Cell Metabolism, 2015 - DOI: 10.1016/j.cmet.2015.02.009
2. Kim S.J. et al. - *Mitochondrially derived peptides as novel regulators of metabolism* - Journal of Physiology, 2017 - DOI: 10.1113/JP274472
3. Reynolds J.C. et al. - *MOTS-c is an exercise-induced mitochondrial-encoded regulator of age-dependent physical decline and muscle homeostasis* - Nature Communications, 2021 - DOI: 10.1038/s41467-020-20790-0

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.